FAKE NEWS DETECTION

PREPARED BY GROUP 2

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INTRODUCTION

BUSINESS UNDERSTANDING

In today's information-driven society, the spread of misinformation and fake news through online news dissemination platforms and social media platforms has become a serious threat to public trust, democracy, health communication, and media integrity. The ability to automatically distinguish fake from real news is essential for:

- Social media platforms
- Fact-checking organizations.
- News aggregators
- The general public.

PROBLEM STATEMENT

The proliferation of fake news on digital platforms poses a significant risk to public trust and decision-making. Traditional methods of identifying fake news are slow and resource-intensive, making it difficult to manage the vast volume of online content.

PROPOSED SOLUTION

In this project we aim to develop an automated binary classification system using transformer-based Natural Language Processing (NLP) models to detect and flag fake news articles with high accuracy. By leveraging advanced NLP techniques, the system will offer a scalable solution to help identify harmful misinformation, improving information credibility and supporting the fight against digital manipulation.

OBJECTIVES

Develop a binary classification system to detect fake news using various models:

- Baseline Logistic Regression Model
- XGBoost Model.
- LSTM Neural Network
- Transfer Learning with a RoBERTa Pretrained Transformer
- Select the best-performing model for deployment.

DATASET

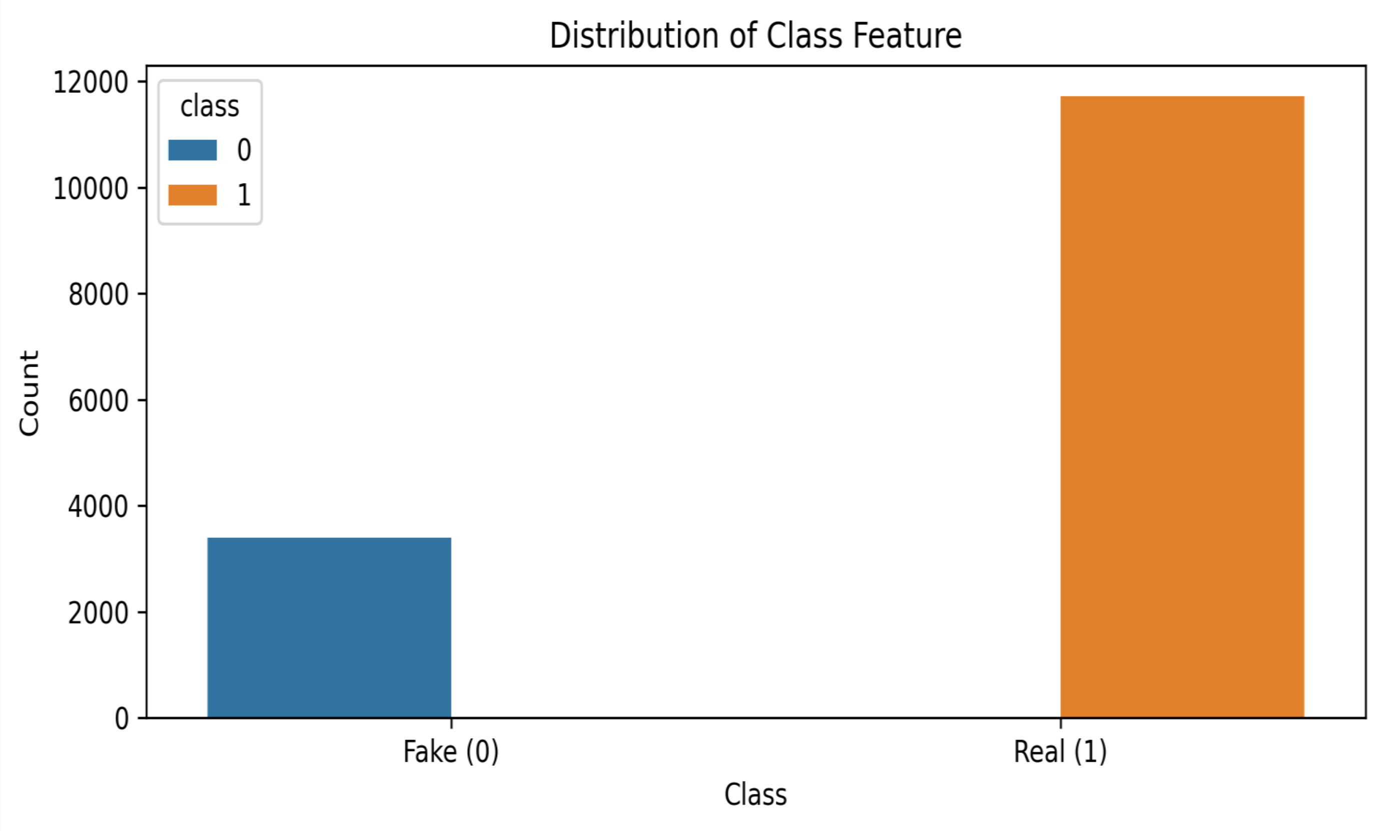
Datasets used;

- ❖ gossipcop_fake.csv
- ❖ gossipcop_real.csv
- ❖ politifact_fake.csv
- ❖ politifact_real.csv

The 4 labelled datasets were cloned from; https://github.com/KaiDMML/FakeNewsNet/tree/master/dataset

Web Scrapping: Formulating a Python script (scraper.py) to scrape text data from the URL link of each entity across the 4 datasets which contains ***fetch_article_text*(URL), *scrape_dataset*** and ***The main*() *function***

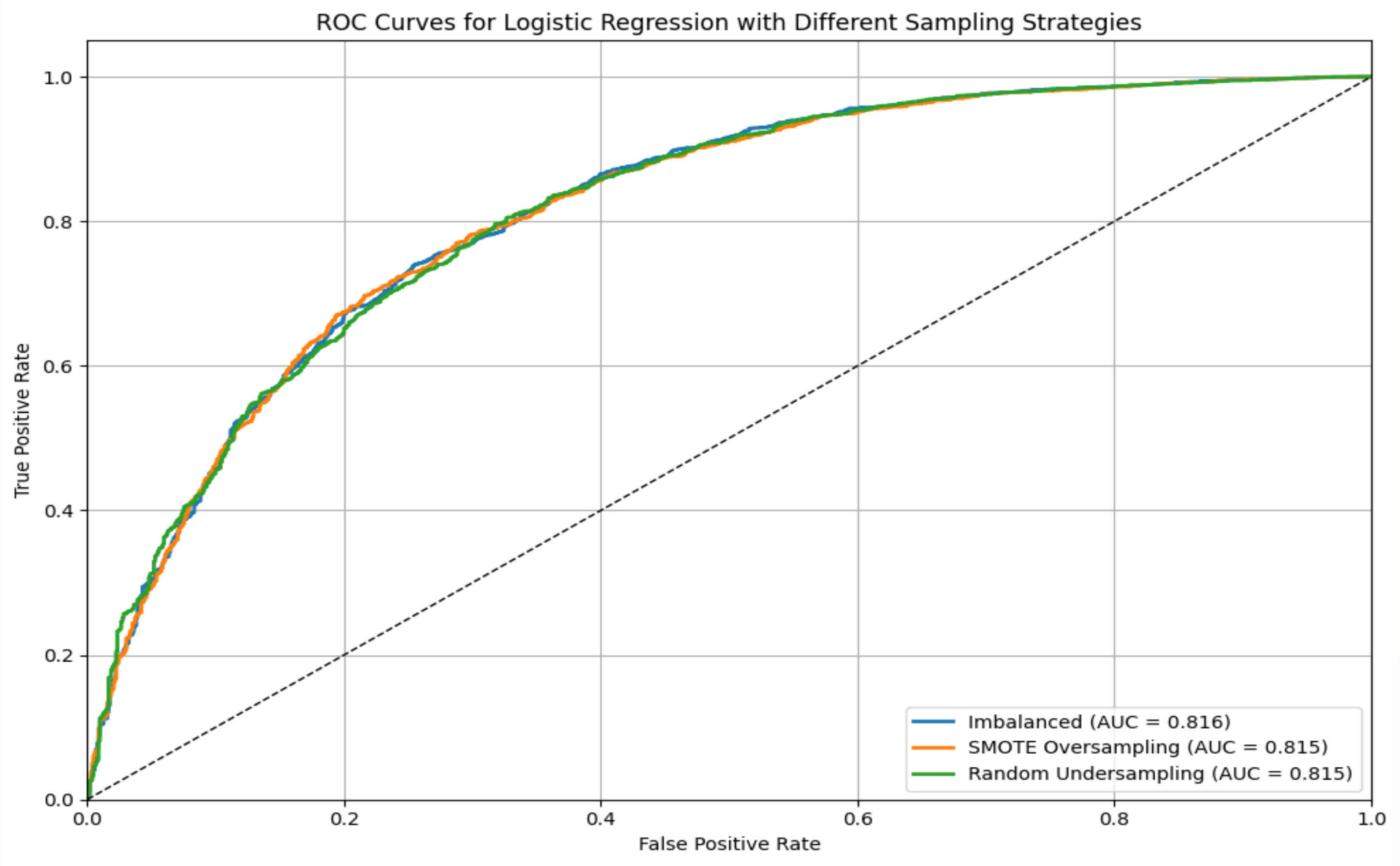
FINDINGS



Class Feature Distribution

Dataset: ~12K real vs. ~3K fake articles (80% real).
For the imbalanced dataset, **SMOTE** was used for balancing the dataset with the finetuned RoBERTa model.

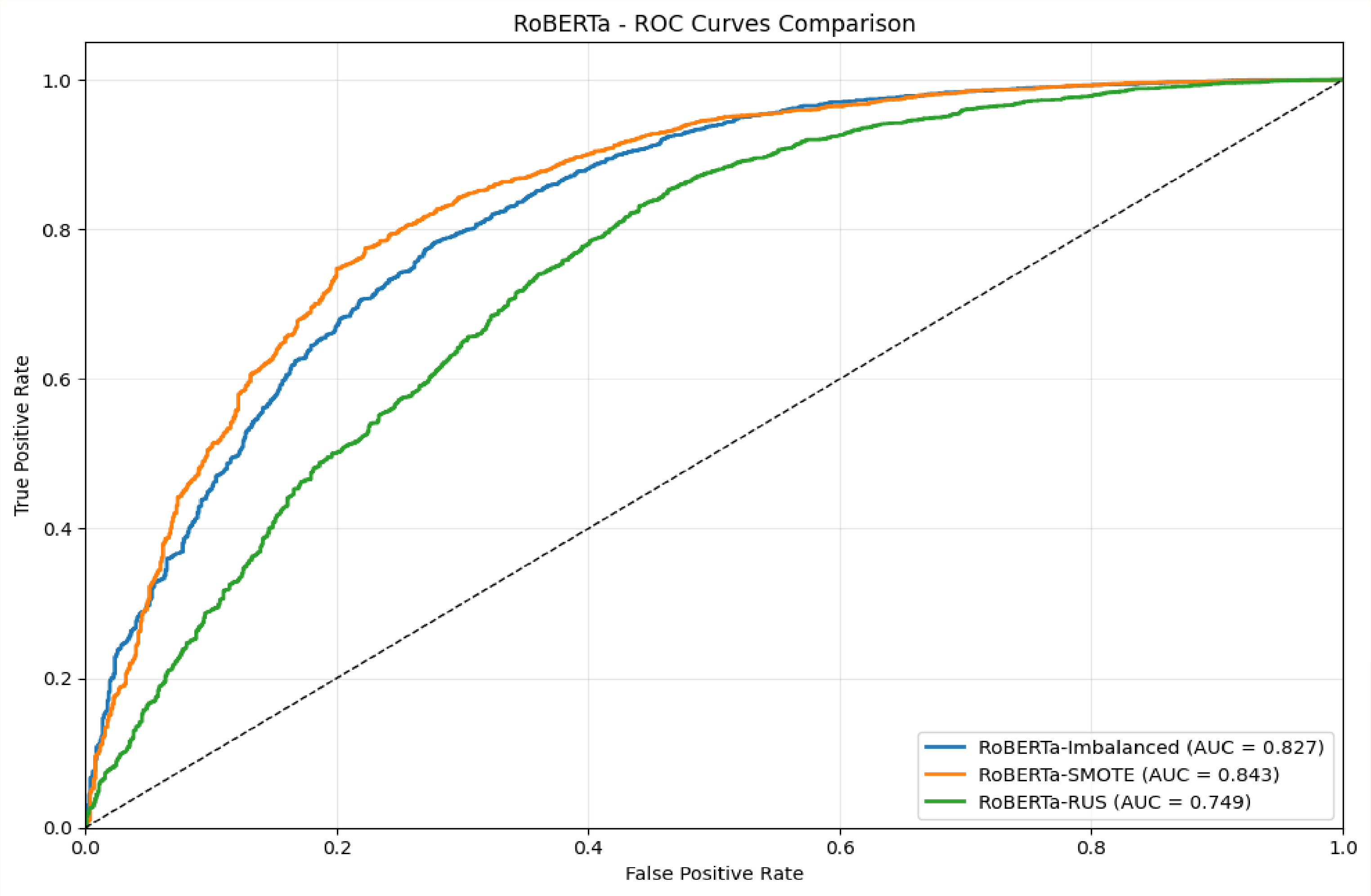
ROC CURVE FOR LOGISTIC REGRESSION



The ***logistic regression model*** achieved *AUC* scores of *0.816* (imbalanced), *0.815* (SMOTE), and 0.815 (under sampling).

The *LSTM* model with *SMOTE*-balanced data achieves the best performance, with the highest macro-averaged recall of 0.7104.

RoBERTa-ROC CURVE ANALYSIS



We evaluated three versions of the RoBERTa-base Transformer.

- RoBERTa-**SMOTE** achieved an AUC Score of 0.843 making it the most effective with enhanced minority class representation, boosting detection accuracy.
- RoBERTa-**Imbalanced** achieved 0.827 Performed well, but showed bias toward majority class.
- RoBERTa-**RUS** achieved 0.749 which is an underperformance.

CONCLUSIONS

The finetuned RoBERTa model with SMOTE balancing achieved the highest macro-averaged recall (0.7660), demonstrating strong performance in detecting fake news. Its pretraining and balanced fine-tuning enhanced recall for the minority class without compromising generalization

RECOMMENDATIONS

- Incorporate more training data especially class 0 (Fake News) entries to improve recall.
- Leverage pretrained embeddings when finetuning the RoBERTa transformer to reduce training time.
- Experiment finetuning other NLP transformers such as DistilBERT and DeBERTa.

NEXT STEPS

- Deploy the containerized finetuned RoBERTa model via AWS SageMaker for low-latency inference.
- Develop browser extensions and social media plugins using the model's API for real-time credibility scoring.
- Incorporate automation frameworks to flag low-confidence predictions for human review.

THANK YOU